

TEF82xx RFE Config Generator

Rev. 0.8.10 — 18 December 2024

User manual

Document information

Information	Content
Keywords	TEF82xx, RFE, Config
Abstract	This user manual describes the rfeConfigGenerator software tool that generates RFE config blobs required by the TEF82xx RFE firmware.



Revision history

Revision history

Rev	Date	Description
0.8.10	20241218	- Several <i>optional</i> parameters have been added, more details here: Section 3. These parameters can be omitted (a default value will be used) and the blob will still generate: - txCalibration (Section 5.1.7) newly added. Configures whether the calibration is done <i>with</i> or <i>without</i> transmission. - txCalibrationThresholds (Section 5.4.5) newly added. Configures high/low thresholds for txCalibration <i>without</i> transmission. - pll-loop-filter-bandwidth (Section 5.5.3) value limits have been removed. The tool will throw a warning instead, when using values that yield higher phase noise, but the blob(s) will still generate. - If using different center-frequency-kHz (Section 5.5.3) in the same <i>sequence</i>, while frequency-drift-steps (Section 5.5.7) is different than 0, the tool will output a warning, stating that only the first center frequency will be used and the rest will be ignored. - frequency-drift-steps (Section 5.5.7) range has been updated, and the values no longer represent a <i>frequency</i>. Instead, they represent a <i>step</i> of 42.9153 Hz. - The tool will output an error, if chirp-interval-time-ticks (Section 5.5.2) is smaller than the sum of its parts: dwell-time-ticks + settle-time-ticks + acquisition-time-ticks + reset-time-ticks (Section 5.5.2) + jumpback Timing (Section 5.1.6). - The tool will output an error, if dwell-time-ticks + settle-time-ticks + reset-time-ticks < 160 (Section 5.5.2). This happens when all 3 parameters are set to minimum value. - Reduced blob size by 33 bytes (fusaFaultMask0 → fusaFaultMask11 sizes in blob reduced from 32bits to 8bits, per set). - Radar Cycle Schedule Generator (Section 6) will generate an error if a minimum idle time of (50μS) is not ensured, for each <i>sequence</i>. - Added details about the structure of Dynamic Table data. (Section 5.6.1) - pdc is now available (Section 5.1.4) - rxSaturationCounter is now available (Section 5.2.1) - CRC has been added to the blob, to ensure data consistency on both M7 cores.
0.8.9	20240705	- fastReset has been removed. - pll-loop-filter-bandwidth (Section 5.5.3) value range has been limited to recommended values. - txEnable has been removed. - Updated ranges for dwell-time-ticks, settle-time-ticks and reset-time-ticks (Section 5.5.2) - Fixed issue where using value 3 for csi2VirtualChannel (Section 5.5.10) would still generate blob files, despite it being reserved for BIST. - Enabling multiple transmitters per profile in the same sequence (see txTransmission Enable (Section 5.5.4), without configuring power mode (Section 5.1.1) to lower power, will throw an error and blob files will not be generated.
0.8.8	20240329	- csi2VirtualChannel 3 (Section 5.5.10) was removed, since it is reserved for BIST. - rxGain (Section 5.5.8) updated with correct values. - profileRepeatCount (Section 5.4.3) newly added. - dynamicTables (Section 5.6.1) are now available. - dynamicUpdate (Section 5.4.4) is now available as a consequence. - Added details on how to build the executable that generates the timing CSV file. (Section 6). - Minor changes - Changed the reference to manual, to the actual name of the file (Section 2).

Revision history...continued

Rev	Date	Description
		– Small description improvements (Section 3), (Section 4) & (Section 5). – Moved XSD file alongside the XML file and removed "resource" folder. – Updated <i>readme.txt</i> file. – Logo updated.
0.8.7	20240229	• <i>txPower</i> (Section 5.5.5) range - lower value, changed from -9 to 3
0.8.6	20240117	• <i>chirpFrequency</i> (Section 5.5.7) moved from <i>chirpSequenceConfigs</i> to <i>chirpProfiles</i>
0.8.5	20240105	• <i>rxPhaseDiff</i> (Section 5.2.7) and <i>rxGainDiff</i> (Section 5.2.8) are now available – Parameter names have been changed to <i>rxPhaseDiffLna</i> and <i>rxGainDiffLna</i>. – 2 new parameters added: <i>rxPhaseDiffMixer</i> and <i>rxGainDiffMixer</i>. • <i>injectTestTone</i> (Section 5.2.11) is now available – Attribute "<i>before-lna</i>" changed to "<i>mode</i>" – Selections have changed to: <i>both lna and mixer</i> <i>only mixer</i> <i>alternate lna mixer</i>. • 3 new Ball Break detector parameters have been added under element <i>checkBBD</i> – <i>loiBbd</i> (Section 5.2.12). – <i>txBbd</i> (Section 5.2.13). – <i>rxBbd</i> (Section 5.2.14). • <i>fuSaFaultMask</i> (Section 5.2.16) is now available – Updated mask names and descriptions • <i>bistInterval</i> (Section 5.3.5) is now available – Added <i>no bist</i> as selection. • <i>chirpFrequency</i> (Section 5.5.7) is now available – Parameter no longer accepts negative values (range adjusted accordingly). • 13 new DDMA parameters have been added under <i>ddma</i> element. (Section 5.4.6) – <i>mode</i> – <i>tx-1-use-ddma</i> – <i>tx-2-use-ddma</i> – <i>tx-3-use-ddma</i> – <i>tx-1-ddma-init-phase</i> – <i>tx-1-ddma-init-phase</i> – <i>tx-1-ddma-init-phase</i> – <i>tx-1-ddma-phase-update</i> – <i>tx-1-ddma-phase-update</i> – <i>tx-1-ddma-phase-update</i> – <i>tx-1-final-pc-gen-mode</i> – <i>tx-1-final-pc-gen-mode</i> – <i>tx-1-final-pc-gen-mode</i> • <i>txPhaseRotation</i> (Section 5.5.6) is now available
0.8.3	20230731	• Added description for cascaded use-case in Section 3 • Updated pll-loop-filter-bandwidth parameter availability in Section 5.5.3 • Updated txEnable parameter availability in txEnable • Removed following parameters: TemperatureThresholdRx, TxPhaseDiff, TxBistPower, TxSelect • Updated version number to match with the release version

1 Introduction

An RFE config blob is required for each TEF82xx, to configure the front-end. The rfeConfigGenerator software tool is provided to convert RFE configurations, from XML format into binary blob data, which can directly be loaded into the memory of the RFE Firmware, using the `CDD_Rfe_Configure()` API.

Note: For cascaded solutions (i.e. more than one front-end devices), multiple blobs must be generated, exactly one for each physical device.

The XML file (`rfeConfig.xml`), is validated through the provided XML Schema file (`rfeConfig_x_y.xsd`), both while editing the file (if using Visual Studio Code, as recommended here: [Section 4](#)) and while generating the blob.

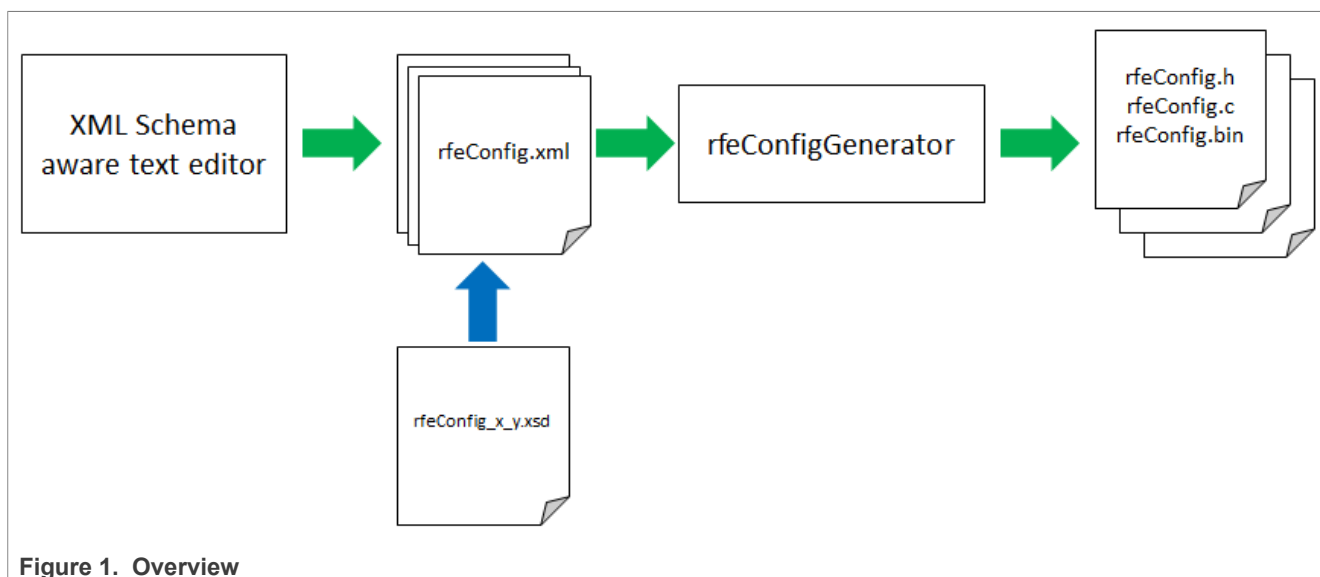


Figure 1. Overview

The directory contents and usage of the rfeConfigGenerator software tool are described in [Section 2](#) and [Section 3](#). The use of the Visual Studio Code editor for writing the rfeConfig.xml file is described in [Section 4](#). The structure of the rfeConfig.xml file itself is described in [Section 5](#).

2 Directory contents

The contents of the directory are listed in [Table 1](#).

Table 1. Directory contents

Item:	Description:
rfeConfigGenerator/	This directory contains the rfeConfigGenerator software tool. A user of this software tool does not need to be concerned with the content of this directory.
readme.txt	Contains a concise description of the rfeConfigGenerator software tool and its usage.
rfeConfig.xml	The user specification of the RFE software config blob.
rfeConfig_tef82xx_x_y.xsd	The XML Schema file that is be used for validating and editing the rfeConfig.xml file. The 'x' and 'y' parts denote the version of the RFE software.
rfeConfigGenerator.bat	A Windows batch file that runs the rfeConfigGenerator software tool.
TEF82xx RFE Config Generator User Manual.pdf	The user manual for the rfeConfigGenerator software tool.

3 Software tool usage

The rfeConfigGenerator software tool can be invoked, by running the `rfeConfigGenerator.bat` batch file with one (or more) `*.xml` files as argument – please see the `readme.txt` file at the same path as this document. The tool will validate each XML file and subsequently parse its content to generate the binary blob(s).

Note:

- A binary file (*.bin) will be generated for each individual *.xml.
- A single *.c file will be generated, that combines all *.xml files.
- XML files are validated independently at generation time. The user is responsible to ensure that all *.xml files are consistently defined, as there is **no cross-checking**.

3.1 Successful generation

When successful, the software tool will generate and overwrite the files listed in [Table 2](#) and prints the following message:

```
Config .h and .c files successfully generated.
Config .bin file successfully generated.
Dynamic tables .h and .c files successfully generated.
Dynamic tables .bin file successfully generated.

Press any key to continue . . .
```

Table 2. Generated files

Item:	Description:
rfeConfig.h and rfeConfig.c	These files contain a representation of the RFE software config blob in the format accepted by the RFE driver. For cascaded use-case, these files will contain a data structure for each XML configuration passed to the generator. (i.e. all configurations will be present in these files)
rfeConfig.bin	This file contains a binary representation of the RFE software config blob that can be used in user environments that are not based on the RFE driver. For cascaded use-case, one separate *.bin file will be generated for each XML configuration passed to the generator.
rfeDynamicTables.h and rfeDynamicTables.c	These files contain a representation of the RFE software dynamic tables blob in the format accepted by the RFE driver. These are generated only when dynamic table(s) are defined in rfeConfig.xml. For cascaded use-case, it will be the same as for the above *.c and *.h files.
rfeDynamicTables.bin	This file contains a binary representation of the RFE software dynamic tables blob that can be used in user environments that are not based on the RFE driver. This is generated only when dynamic table(s) are defined in rfeConfig.xml. For cascaded use-case, it will be the same as for the above *.bin files.

3.2 Failure

When a failure is detected by the XML Schema validator during the validation of the `rfeConfig.xml` file, the software tool will print a message similar to:

```
Error: cvc-pattern-valid: Value '1 MHz' is not facet-valid with respect to pattern '40 MHz|20 MHz|10 MHz|5 MHz' for type 'rfeApi_effectiveSamplingFrequency_t'.
```

```
Error: cvc-attribute.3: The value '1 MHz' of attribute 'frequency' on
element 'effectiveSamplingFrequency' is not valid with respect to its type,
'rfeApi_effectiveSamplingFrequency_t'.
Error: unable to validate the XML file!
Error: Config parsing failed!

Press any key to continue . . .
```

This message indicates that the `frequency` attribute of the `effectiveSamplingFrequency` is invalid as 1 MHz. The user should edit the `rfeConfig.xml` file and change this attribute value such that it is one of valid values: 40 MHz, 20 MHz, 10 MHz, 5 MHz.

When a failure is detected during parsing of the `rfeConfig.xml` file, the software tool will print a message similar to:

```
Error detected when parsing the XML Config file:
Error: unable to parse the radarCycleConfig XML Element!
Unable to parse recalibrationProfileDependent element!
Error: Config parsing failed!

Press any key to continue . . .
```

This message indicates that an `recalibrationProfileDependent` element is causing issues. In this case, there is a duplicated `chirp-sequence` attribute value that should be resolved.

3.3 Optional Parameters

Starting with version 0.8.10, the `rfeConfig.xml`, also has *optional* parameters. These parameters can be omitted from the XML, and the generation process will still be successful → the blob(s) will be created. For this reason, these parameters also have a *default* value so that they do not cause unknown behavior. These values are generally an "OFF" state for functionalities or a set of NXP defined values for thresholds, levels etc. These parameters have the "(optional)" keyword next to them, as well as the "(default)" keyword, next to the default selection:

5.1.7 txCalibration (optional)

The `txCalibration` element configures whether the T antennas radiating normally or with reduced radiation a

Note: *If this is set to without transmission, then t be configured.*

Attribute mode	
Configures how TX calibration shall be performed.	
without transmission 	TX calibration shall be p antenna radiation will be
with transmission (default)	TX calibration shall be p normally with this setting

4 XML file editing

The Visual Studio Code editor¹ — in combination with the "XML Language Support" extension offered by Red Hat² — is advised to edit the `rfeConfig.xml` file.

The code block below shows the minimal content required to enable the XML Schema aware text editor to validate a `rfeConfig.xml` file. This is done by referencing the schema (`xsi:noNamespaceSchemaLocation="rfeConfig_tef82xx_0.8.10.0.xsd"`) provided in the same folder as this manual:

```
<?xml version="1.0"?>
<rfeConfig xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:noNamespaceSchemaLocation="rfeConfig_tef82xx_0.8.10.0.xsd">
</rfeConfig>
```

When creating the XML file, Visual Studio Code editor will indicate if the XML is invalid (Figure 2). The editor underscores the root element with a red squiggly line to indicate that a child element is missing (upper red arrow). The error counter in the lower left corner of the window (lower red arrow) also indicates that errors are present. This counter can be clicked to toggle the visibility of the detailed error panel. The detailed error message indicates that the `generalConfig` child element is missing (middle red arrow). The detailed error message indicates that the `generalConfig` child element is missing (middle red arrow).

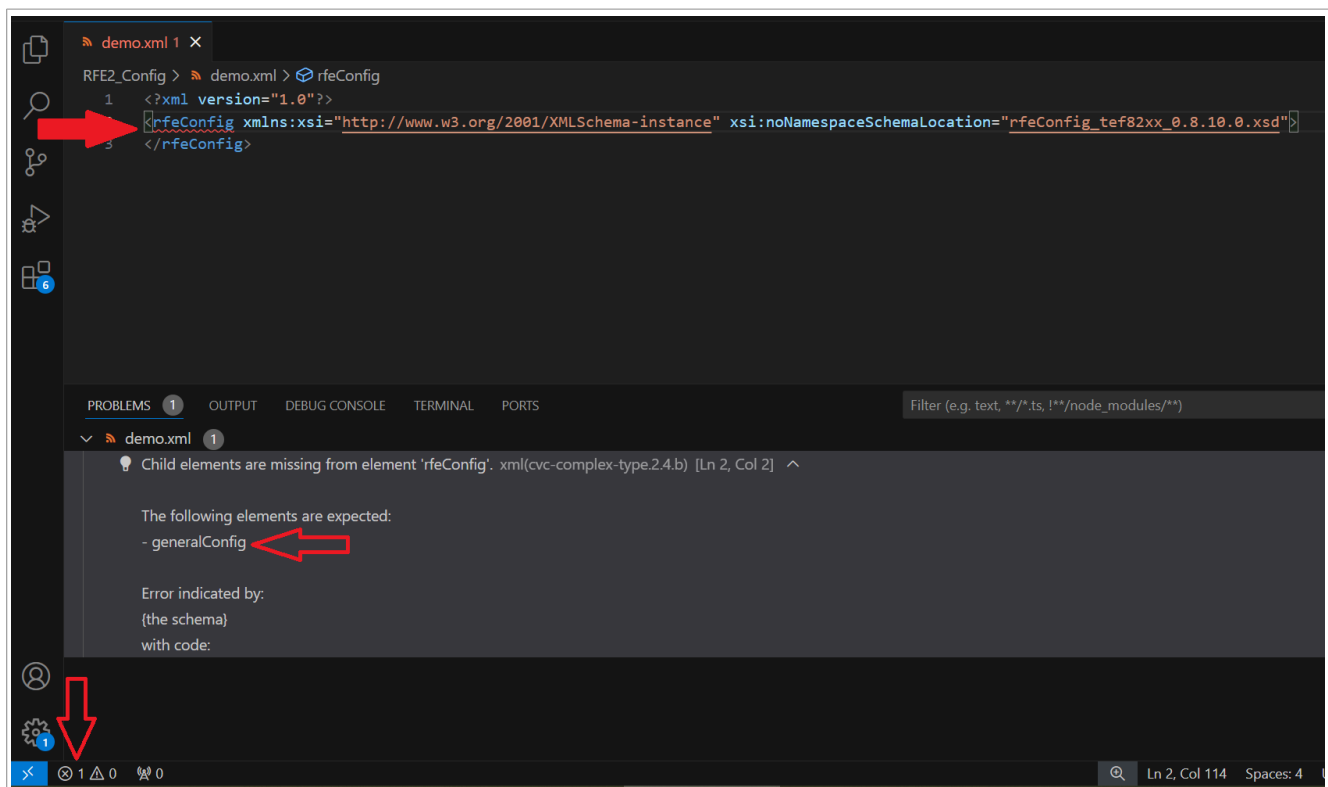


Figure 2. Minimal document

Typing '`<`' will make the editor display a suggestion list with all valid items that could be inserted at that point. In the case of Figure 3, a `generalConfig` element can be added.

¹ <https://code.visualstudio.com/>
² <https://marketplace.visualstudio.com/items?itemName=redhat.vscode-xml>

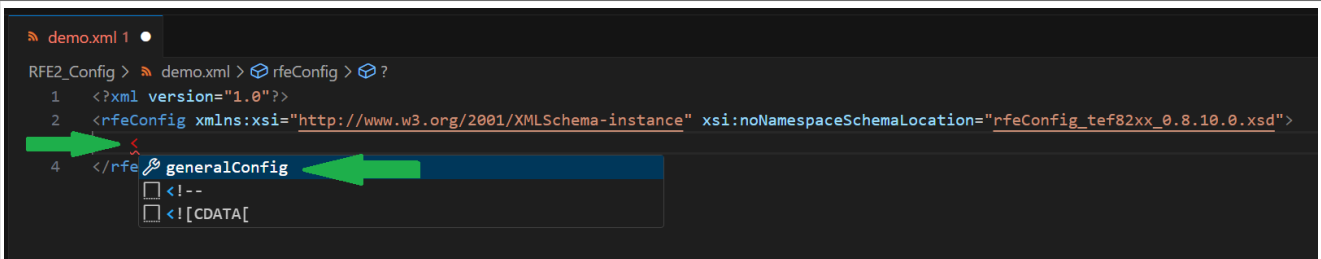


Figure 3. Element insertion

The editor will also automatically add all mandatory attributes for an element when this element is added to the XML structure as shown in Figure 4. The green arrows in this figure indicate the two attributes defined for the ioSignals element and the red arrows indicate the empty values added for these attributes that should be filled in by the user.

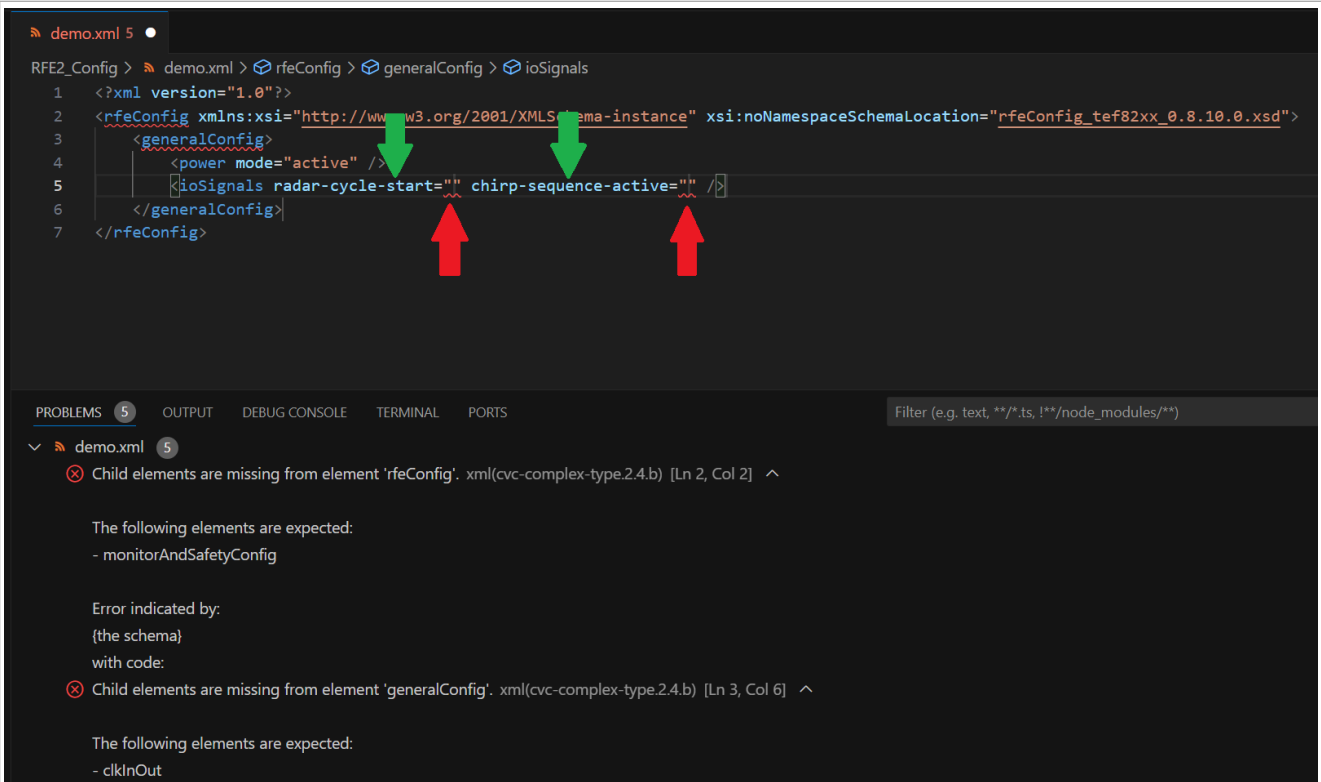


Figure 4. Element attributes

When hovering over an invalid attribute value, the editor will display a panel indicating why this value is invalid. In the case of Figure 5, the editor indicates that there are two valid values for the mode attribute: lower power or active (yellow arrows).

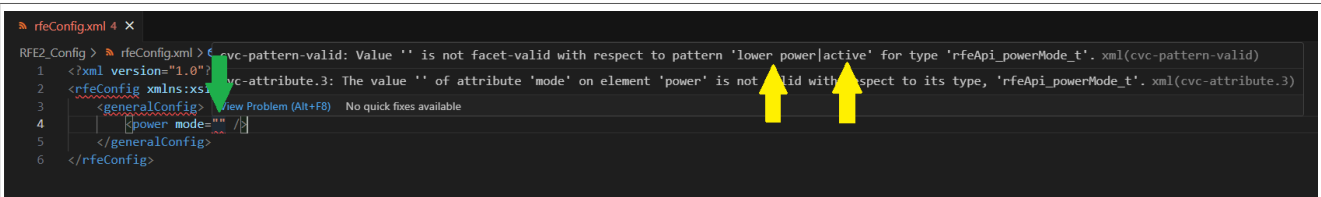


Figure 5. Element attribute values

The error counter will indicate 0 errors when the rfeConfig.xml document is deemed valid by the XML Schema validator as shown in [Figure 6](#).

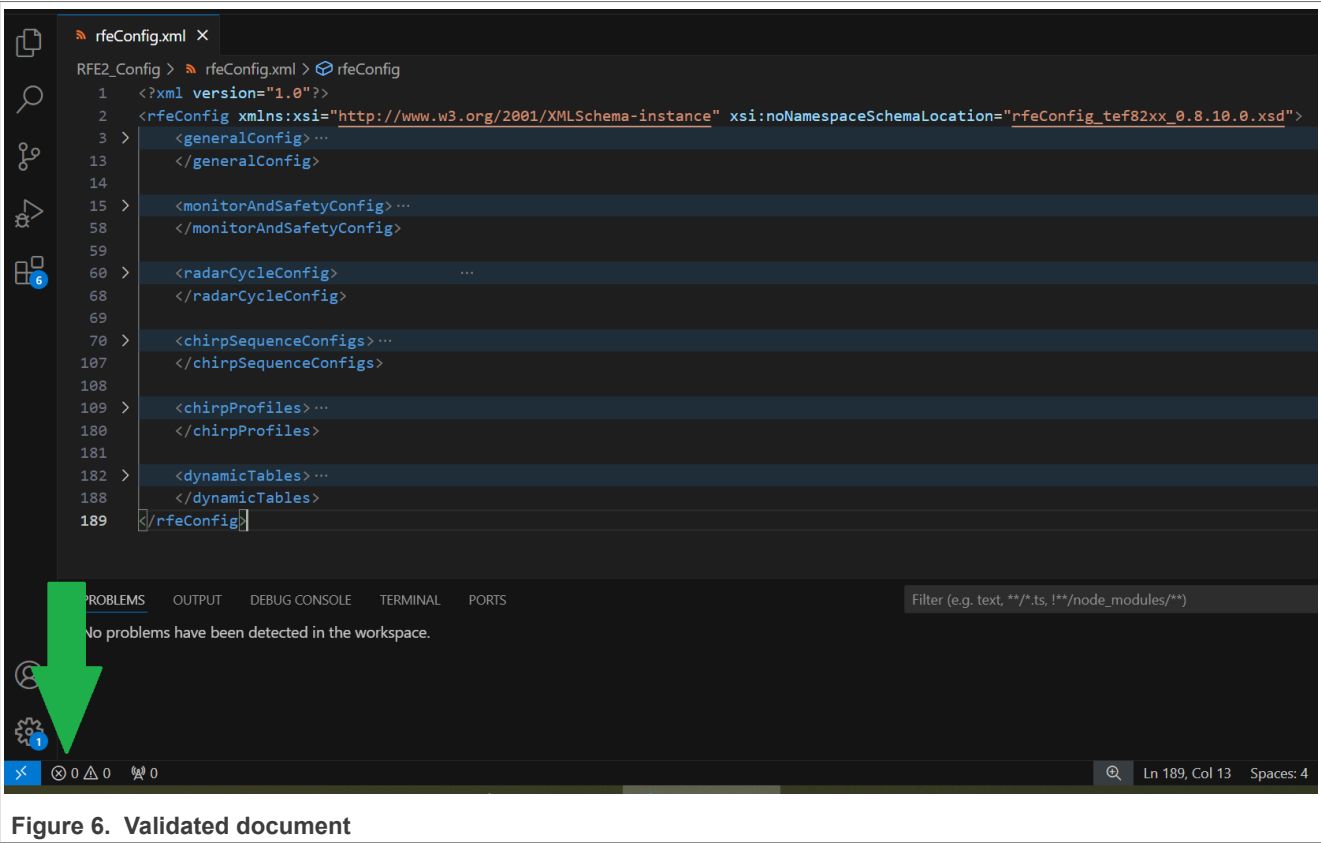


Figure 6. Validated document

5 XML structure

This chapter describes the structure of the `rfeConfig.xml` file.

Please refer to the RF Abstract 2.0 API manual (RSDK User Manual->Software Modules->RFE Abstract 2.0 API) and to the TEF82XX Reference Manual (RM00227) for further details on the specific parameters specified in the RFE config blob.

The root element must be named `rfeConfig` and refer to a valid XSD version such as `rfeConfig_tef82xx_0.8.10.0.xsd`. The version of the XSD will dictate the available parameters (and their allowed value range). The XML root element must look as below code block:

```
<?xml version="1.0"?>
<rfeConfig xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:noNamespaceSchemaLocation="rfeConfig_tef82xx_0.8.10.0.xsd">
  ...
</rfeConfig>
```

The top-level structure of the `rfeConfig.xml` file is:

Element:	Occurence:
<rfeConfig ...>	1
<generalConfig>	1
<monitorAndSafetyConfig>	1
<radarCycleConfig>	1
<chirpSequenceConfigs>	1
<chirpProfiles>	1
<dynamicTables>	0 - 1
</rfeConfig>	

The remainder of this chapter details these top-level elements.

Note: *Keep in mind that the elements marked with "(not available)" are still present in the XML configuration file, but they will not produce any changes in the behavior of the Front End.*

5.1 generalConfig

The structure of the `generalConfig` element is:

Element:	Occurence:
<generalConfig>	1
<power ...>	1
<ioSignals ...>	1
<clkInOut ...>	1
<pdc ...>	1
<dataOut ...>	1
<metadata ...>	1
</dataOut>	

Element:	Occurence:
<jumpbackTiming ...>	1
<txCalibration ...>	1
</generalConfig>	

5.1.1 power

The `power` element configures the power mode of the RFE.

Attribute <code>mode</code>	
Selects the power mode for the RFE. If a sequence uses multiple profiles with different TXs enabled (see <i>txTransmissionEnable</i> (Section 5.5.4)), this setting must be configured to "lower power" otherwise a lock-step error would be triggered from ISM. To prevent this, the generator will not generate the blob files if "lower power" is not used in the described scenario, and it will display the error message: <i>"Attempting to use different TX antennas per profile, without setting <power mode= \"lower power\">"</i>	
<code>active</code>	Puts RFE in active power mode, where all IPs are powered up.
<code>lower power</code>	Puts RFE in a lower power mode, where unused IPs are put in power-saving state during non-chirping activities.

5.1.2 ioSignals (not available)

The `ioSginals` element configures the GPIO of the RFE.

Attribute <code>radar-cycle-start</code>			
Selects the RFE GPIO to route the radar cycle start signal to. The selected RFE GPIO will output an high pulse at the start of every radar cycle.			
	RFE signal:	Pin name:	BGA ball:
<code>gpio 0</code>	DEBUG_RFE0	GPIO_5	W5
<code>gpio 1</code>	DEBUG_RFE1	GPIO_12	V19
<code>gpio 2</code>	DEBUG_RFE2	GPIO_13	V20
<code>gpio 3</code>	DEBUG_RFE3	GPIO_14	W20
<code>gpio 4</code>	DEBUG_RFE4	GPIO_15	W19
<code>gpio 5</code>	DEBUG_RFE5	GPIO_18	N20
<code>gpio 6</code>	DEBUG_RFE6	GPIO_19	T20
<code>gpio 7</code>	DEBUG_RFE7	GPIO_20	R20
<code>disabled</code>	The signal is disabled.		

Attribute <code>chirp-sequence-active</code>			
Selects the RFE GPIO to route the chirp sequence active signal to. The selected RFE GPIO output will be asserted during the chirp sequence and deasserted when a chirp sequence is not active.			
	RFE signal:	Pin name:	BGA ball:
<code>gpio 0</code>	DEBUG_RFE0	GPIO_5	W5
<code>gpio 1</code>	DEBUG_RFE1	GPIO_12	V19

Attribute chirp-sequence-active			
gpio 2	DEBUG_RFE2	GPIO_13	V20
gpio 3	DEBUG_RFE3	GPIO_14	W20
gpio 4	DEBUG_RFE4	GPIO_15	W19
gpio 5	DEBUG_RFE5	GPIO_18	N20
gpio 6	DEBUG_RFE6	GPIO_19	T20
gpio 7	DEBUG_RFE7	GPIO_20	R20
disabled	The signal is disabled.		

5.1.3 clkInOut (not available)

The `clkInOut` element configures the CLK_IN / CLK_OUT IO parameters.

Attribute in	
Disables or enables the CLK_IN IO.	
disabled	The CLK_IN IO is disabled.
enabled	The CLK_IN IO is enabled.

Attribute out	
Disables or enables the CLK_OUT IO.	
disabled	The CLK_OUT IO is disabled.
enabled	The CLK_OUT IO is enabled.

Attribute reduce-driver-level	
Disables or enables the CLK_OUT IO drive level reduction.	
false	The CLK_OUT IO drive level is normal.
true	The CLK_OUT IO drive level is reduced.

5.1.4 pdc

The `pdc` element configures the programmable decimation chain of the RFE. For a detailed description, please refer to the Reference Manual (RM00227), chapter "Programmable decimation chain".

Attribute decimation-filter	
Selects the decimation filter of the programmable decimation chain. This option reduces the effective sampling rate of the ADC as follows:	
narrow	Narrow bandwidth with a passband response up to $0.4 \times f_{\text{EFFECTIVE_SAMPLE_RATE}}$.
wide	Wide bandwidth with a passband response up to $0.5 \times f_{\text{EFFECTIVE_SAMPLE_RATE}}$.

5.1.5 metadata (not available)

The `metadata` element configures the enables metadata long packet for output data.

Attribute packet	
Disables or enables the metadata long packet for output data.	
disabled	The metadata long packet is disabled.
enabled	The metadata long packet is enabled.

Attribute hardware-field	
Disables or enables the hardware fields in the metadata long packet for output data.	
disabled	The hardware fields are disabled.
enabled	The hardware fields are enabled.

Attribute software-content	
Determines what content will be written to the software fields of the metadata long packet.	
rx stage 1 in-phase saturation count	The 4 software fields of the metadata packet contain the stage 1 in-phase saturation count of the 4 receivers.
rx stage 1 quadrature saturation count	The 4 software fields of the metadata packet contain the stage 1 quadrature saturation count of the 4 receivers.
rx stage 2 in-phase saturation count	The 4 software fields of the metadata packet contain the stage 2 in-phase saturation count of the 4 receivers.
rx stage 2 quadrature saturation count	The 4 software fields of the metadata packet contain the stage 2 quadrature saturation count of the 4 receivers.
chirp sequence timestamp	The first software field of the metadata packet contains the RFE timestamp of the start of dwell time of the first chirp in the sequence.

Attribute virtual-channel	
The virtual channel applied to the metadata long packet.	
0	Virtual channel 0 is selected.
1	Virtual channel 1 is selected.
2	Virtual channel 2 is selected.
3	Virtual channel 3 is selected.

5.1.6 jumpbackTiming

The jumpbackTiming element configures the jumpback timing.

Attribute time-ticks	
Sets the chirp jumpback time in steps of 25 nanoseconds.	
1 - 254	Allowed range.

5.1.7 txCalibration (optional)

The txCalibration element configures whether the TX calibration procedure will be performed with the TX antennas **radiating** normally or with reduced radiation amplitude.

Note: If this is set to *without transmission*, then txCalibrationThresholds (Section 5.4.5) must also be configured.

Attribute mode	
Configures how TX calibration shall be performed.	
without transmission	TX calibration shall be performed with reduced TX radiation amplitude (i.e. TX antenna radiation will be significantly reduced)
with transmission (default)	TX calibration shall be performed with transmission (i.e. TX antenna will be radiating normally with this setting)

5.2 monitorAndSafetyConfig

The structure of the monitorAndSafetyConfig element is:

Element:	Occurence:
<monitorAndSafetyConfig>	1
<rxSaturation ...>	1
<rxSaturationCounter ...>	1
<rxSaturationThresholds>	1
<rxSaturationThresholdRx1 ...>	1
<rxSaturationThresholdRx2 ...>	1
<rxSaturationThresholdRx3 ...>	1
<rxSaturationThresholdRx4 ...>	1
</rxSaturationThresholds>	
</rxSaturation>	
<adcClippingCounter ...>	1
<temperatureThresholds>	1
<temperatureThresholdTx12 ...>	1
<temperatureThresholdTx3 ...>	1
<temperatureThresholdXo ...>	1
</temperatureThresholds>	
<r1FaultPromotion ...>	1

Element:	Occurence:
<frequencyForBist ...>	1
<rxBist>	1
<rxPhaseDiffLna ...>	1
<rxGainDiffLna ...>	1
<rxPhaseDiffMixer ...>	1
<rxGainDiffMixer ...>	1
<injectTestTone ...>	1
</rxBist>	
<checkBBD>	1
<loiBbd ...>	1
<txBbd ...>	1
<rxBbd ...>	1
</checkBBD>	
<txPpdThreshold ...>	1
<fuSaFaultMask0 ...>	1
<fuSaFaultMask1 ...>	1
<fuSaFaultMask2 ...>	1
<fuSaFaultMask3 ...>	1
<fuSaFaultMask4 ...>	1
<fuSaFaultMask5 ...>	1
<fuSaFaultMask6 ...>	1
<fuSaFaultMask7 ...>	1
<fuSaFaultMask8 ...>	1
<fuSaFaultMask9 ...>	1
<fuSaFaultMask10 ...>	1
<fuSaFaultMask11 ...>	1
</monitorAndSafetyConfig>	

5.2.1 rxSaturationCounter

The rxSaturationCounter element configures the resetting of the receiver saturation counts.

Attribute reset-every-sequence	
Disables or enables resetting of the receiver saturation counts every chirp sequence.	
false	Resetting of the receiver saturation counts every chirp sequence is disabled.
true	Resetting of the receiver saturation counts every chirp sequence is enabled.

5.2.2 rxSaturationThresholds

The rxSaturationThresholds element configures the saturation values for each receiver. Each RX antenna can be configured individually, through the following sub-elements:

- rxSaturationThresholdRx1
- rxSaturationThresholdRx2
- rxSaturationThresholdRx3
- rxSaturationThresholdRx4

Attribute power-stage-1	
The power threshold that a sample must exceed in stage 1 to be counted as saturated.	
-13.05 dB	RX saturation level is -13.05 dB.
-12.18 dB	RX saturation level is -12.18 dB.
-11.31 dB	RX saturation level is -11.31 dB.
-10.44 dB	RX saturation level is -10.44 dB.
-9.57 dB	RX saturation level is -9.57 dB.
-8.70 dB	RX saturation level is -8.70 dB.
-7.83 dB	RX saturation level is -7.83 dB.
-6.96 dB	RX saturation level is -6.96 dB.
-6.09 dB	RX saturation level is -6.09 dB.
-5.22 dB	RX saturation level is -5.22 dB.
-4.35 dB	RX saturation level is -4.35 dB.
-3.48 dB	RX saturation level is -3.48 dB.
-2.61 dB	RX saturation level is -2.61 dB.
-1.74 dB	RX saturation level is -1.74 dB.
-0.87 dB	RX saturation level is -0.87 dB.
0 dB	RX saturation level is 0 dB.

Attribute max-count-stage-1	
The number of saturations in stage 1 that raises an error.	
0 - 4194303	Allowed range.

Attribute power-stage-2 (not available)	
The power threshold a sample must exceed in stage 2 to be counted as saturated.	
-13.05 dB	RX saturation level is -13.05 dB.
-12.18 dB	RX saturation level is -12.18 dB.
-11.31 dB	RX saturation level is -11.31 dB.
-10.44 dB	RX saturation level is -10.44 dB.
-9.57 dB	RX saturation level is -9.57 dB.

Attribute power-stage-2 (not available)

-8.70 dB	RX saturation level is -8.70 dB.
-7.83 dB	RX saturation level is -7.83 dB.
-6.96 dB	RX saturation level is -6.96 dB.
-6.09 dB	RX saturation level is -6.09 dB.
-5.22 dB	RX saturation level is -5.22 dB.
-4.35 dB	RX saturation level is -4.35 dB.
-3.48 dB	RX saturation level is -3.48 dB.
-2.61 dB	RX saturation level is -2.61 dB.
-1.74 dB	RX saturation level is -1.74 dB.
-0.87 dB	RX saturation level is -0.87 dB.
0 dB	RX saturation level is 0 dB.

Attribute max-count-stage-2 (not available)

The number of saturations in stage 2 that raises an error.

0 - 4194303 Allowed range.

5.2.3 adcClippingCounter

The `adcClippingCounter` element configures the power mode of the RFE.

Attribute reset-every-sequence

Disables or enables the reset of the programmable decimation chain clipping counter, every chirp sequence.

false	Resetting of the programmable decimation chain clipping counts every chirp sequence is disabled.
true	Resetting of the programmable decimation chain clipping counts every chirp sequence is enabled.

Attribute limit-adc1 (not available)

The number of ADC clipping in receiver 1 that raises a warning.

0 - 4194303 Allowed range.

Attribute limit-adc2 (not available)

The number of ADC clipping in receiver 2 that raises a warning.

0 - 4194303 Allowed range.

Attribute limit-adc3 (not available)

The number of ADC clipping in receiver 3 that raises a warning.

0 - 4194303 Allowed range.

Attribute limit-adc4 (not available)

The number of ADC clipping in receiver 4 that raises a warning.

0 - 4194303 Allowed range.

5.2.4 temperatureThresholds (not available)

The `temperatureThresholds` element configures the thresholds for the temperature monitoring sensor of TX1-2 pair, TX3 and Chirp PLL. Same settings, apply to all. The element is comprised of the following sub-elements:

- `temperatureThresholdTx12`
- `temperatureThresholdTx3`
- `temperatureThresholdChirpPll`

Attribute under-error-celsius

Sets the threshold that triggers the under-temperature error.

-40 - 150 Temperature threshold in degrees Celsius.

Attribute high-warning-celsius

Sets the threshold that triggers the high-temperature error.

-40 - 150 Temperature threshold in degrees Celsius.

Attribute over-error-celsius

Sets the threshold that triggers the over-temperature error.

-40 - 150 Temperature threshold in degrees Celsius.

5.2.5 r1FaultPromotion (not available)

The `r1FaultPromotion` element configures R1 fault promotion

Attribute threshold-value

The threshold value is used by RFE FW to classify the R1 fault as R2 fault.

0 - 255 Allowed range. To treat R1 fault as R2 fault by RFE FW, set the threshold value as 0.

5.2.6 frequencyForBist

The `frequencyForBist` element configures the frequency band to be used for RX BIST.

Attribute <code>frequency-in-kHz</code>	
The frequency band to be used for BIST, expressed in kHz.	
75000000 - 82000000	Allowed range.

5.2.7 rxPhaseDiffLna

The `rxPhaseDiffLna` element configures the Rx phase difference (for LNA use-case) which will be used to compare with measured values during RX BIST.

Example:

1500 (XML) = 150.0 (Front-end)

1354 (XML) = 135.4 (Front-end)

400 (XML) = 40.0 (Front-end)

Attribute <code>zero-hr-ref-rx12</code>	
The reference RX phase difference for rx1-rx2 represented in steps of 0.1 degree.	
-1800 - 1800	Allowed range.

Attribute <code>zero-hr-ref-rx13</code>	
The reference RX phase difference for rx1-rx3 represented in steps of 0.1 degree.	
-1800 - 1800	Allowed range.

Attribute <code>zero-hr-ref-rx14</code>	
The reference RX phase difference for rx1-rx4 represented in steps of 0.1 degree.	
-1800 - 1800	Allowed range.

Attribute <code>phase-diff-threshold-tolerance</code>	
The difference in measured RX to RX phase difference with the zero hour threshold, that is allowed, for zero-hr-ref-rx12, zero-hr-ref-rx13 and zero-hr-ref-rx14, represented in steps of 0.1 degree.	
0 - 400	Allowed range.

5.2.8 rxGainDiffLna

The `rxGainDiff` element configures the Rx gain difference (for LNA use-case) which will be used to compare with measured values during RX BIST.

Example:

300 (XML) = 30.0 (Front-end)

157 (XML) = 15.7 (Front-end)

Attribute <code>zero-hr-ref-rx12</code>	
The reference RX gain difference for rx1-rx2 represented in steps of 0.1 dB.	
-400 - 400	Allowed range.

Attribute zero-hr-ref-rx13

The reference RX gain difference for rx1-rx3 represented in steps of 0.1 dB.

-400 - 400 Allowed range.

Attribute zero-hr-ref-rx14

The reference RX gain difference for rx1-rx4 represented in steps of 0.1 dB.

-400 - 400 Allowed range.

Attribute gain-diff-threshold-tolerance

The difference in measured RX to RX gain difference with the zero hour threshold, that is allowed for zero-hr-ref-rx12, zero-hr-ref-rx13 and zero-hr-ref-rx14, represented in steps of 0.1 dB.

0 - 400 Allowed range.

5.2.9 rxPhaseDiffMixer

The `rxPhaseDiffMixer` element configures the Rx phase difference (for Mixer use-case) which will be used to compare with measured values during RX BIST.

Example:

1500 (XML) = 150.0 (Front-end)

1354 (XML) = 135.4 (Front-end)

400 (XML) = 40.0 (Front-end)

Attribute zero-hr-ref-rx12

The reference RX phase difference for rx1-rx2 represented in steps of 0.1 degree.

-1800 - 1800 Allowed range.

Attribute zero-hr-ref-rx13

The reference RX phase difference for rx1-rx3 represented in steps of 0.1 degree.

-1800 - 1800 Allowed range.

Attribute zero-hr-ref-rx14

The reference RX phase difference for rx1-rx4 represented in steps of 0.1 degree.

-1800 - 1800 Allowed range.

Attribute phase-diff-threshold-tolerance

The difference in measured RX to RX phase difference with the zero hour threshold, that is allowed, for zero-hr-ref-rx12, zero-hr-ref-rx13 and zero-hr-ref-rx14, represented in steps of 0.1 degree.

Attribute <code>phase-diff-threshold-tolerance</code>	
0 - 400	Allowed range.

5.2.10 rxGainDiffMixer

The `rxGainDiffMixer` element configures the Rx gain difference (for Mixer use-case) which will be used to compare with measured values during RX BIST.

Example:

300 (XML) = 30.0 (Front-end)
157 (XML) = 15.7 (Front-end)

Attribute <code>zero-hr-ref-rx12</code>	
The reference RX gain difference for rx1-rx2 represented in steps of 0.1 degree.	
-400 - 400	Allowed range.

Attribute <code>zero-hr-ref-rx13</code>	
The reference RX gain difference for rx1-rx3 represented in steps of 0.1 degree.	
-400 - 400	Allowed range.

Attribute <code>zero-hr-ref-rx14</code>	
The reference RX gain difference for rx1-rx4 represented in steps of 0.1 degree.	
-400 - 400	Allowed range.

Attribute <code>gain-diff-threshold-tolerance</code>	
The difference in measured RX to RX gain difference with the zero hour threshold, that is allowed for <code>zero-hr-ref-rx12</code> , <code>zero-hr-ref-rx13</code> and <code>zero-hr-ref-rx14</code> , represented in steps of 0.1 degree.	
0 - 400	Allowed range.

5.2.11 injectTestTone

The `injectTestTone` element configures where the application of the test tone is performed.

Attribute <code>mode</code>	
The test tone for RX BIST can be injected before LNA or after LNA (i.e. before Mixer).	
<code>both lna and mixer</code>	Test tone is applied both before the LNA & the Mixer, every RX BIST sequence.
<code>only mixer</code>	Test tone is applied only before Mixer, every RX BIST sequence.
<code>alternate lna mixer</code>	Test tone is applied alternatively for each RX BIST sequence: one sequence before LNA, the next sequence before Mixer and so on.

5.2.12 loiBbd

The `loiBbd` element configures the Ball Break detector settings related to the LO interface.

Attribute <code>loi-bbd-enable</code>	
Enables or disables the Ball Break detector for LO interface.	
<code>disabled</code>	Disables the Ball Break detector for LO Interface.
<code>enabled</code>	Enables the Ball Break detector for LO Interface.

Attribute <code>loout-sensitivity</code>	
Sets the current sensitivity for the LO-out Ball Break detector.	
<code>normal</code>	Normal sensitivity (50 μ A).
<code>high</code>	High sensitivity for long differential antenna feed-lines (500 μ A).

Attribute <code>loout-grounded</code>	
Configures whether the Ball Break detector balun is grounded or not, for the LO-out.	
<code>not grounded</code>	Balun not grounded.
<code>grounded</code>	Balun grounded.

Attribute <code>loin-sensitivity</code>	
Sets the current sensitivity for the LO-in Ball Break detector.	
<code>normal</code>	Normal sensitivity (50 μ A)
<code>high</code>	High sensitivity (500 μ A)

Attribute <code>loin-grounded</code>	
Configures whether the Ball Break detector balun is grounded or not, for the LO-in.	
<code>not grounded</code>	Balun not grounded.
<code>grounded</code>	Balun grounded.

5.2.13 txBbd

The `txBbd` element configures the Ball Break detector settings related to the TX antennas.

Attribute <code>tx-bbd-enable</code>	
Enables or disables the Ball Break detector for the TX antennas.	
<code>disabled</code>	Ball Break detector disabled.
<code>enabled</code>	Ball Break detector enabled.

Attribute tx-sensitivity	
Sets the current sensitivity for the TX antennas Ball Break detector.	
normal	Normal sensitivity (50µA).
high	High sensitivity (500µA).

Attribute tx-grounded	
Configures whether the Ball Break detector balun is grounded or not, for the TX antennas.	
not grounded	Balun not grounded.
grounded	Balun grounded.

5.2.14 rxBbd

The `rxBbd` element configures the Ball Break detector settings related to the RX antennas.

Attribute rx-bbd-enable	
Enables or disables the Ball Break detector for the RX antennas.	
disabled	Ball Break detector disabled.
enabled	Ball Break detector enabled.

Attribute rx-sensitivity	
Sets the current sensitivity for the RX antennas Ball Break detector.	
normal	Normal sensitivity (50µA).
high	High sensitivity (500µA).

Attribute rx-grounded	
Configures whether the Ball Break detector balun is grounded or not, for the RX antennas.	
not grounded	Balun not grounded.
grounded	Balun grounded.

5.2.15 txPpdThreshold (not available)

The `txPpdThreshold` element configures the threshold power level for peak power detector .

Attribute profile0-dBm	
The threshold power level of peak power detector for profile0 in the steps of 0.1 dBm.	
-9 - 15	Allowed range

Attribute <code>profile1-dBm</code>	
The threshold power level of peak power detector for profile1 in the steps of 0.1 dBm.	
-9 - 15	Allowed range

Attribute <code>profile2-dBm</code>	
The threshold power level of peak power detector for profile2 in the steps of 0.1 dBm.	
-9 - 15	Allowed range

Attribute <code>profile3-dBm</code>	
The threshold power level of peak power detector for profile3 in the steps of 0.1 dBm.	
-9 - 15	Allowed range

Attribute <code>profile4-dBm</code>	
The threshold power level of peak power detector for profile4 in the steps of 0.1 dBm.	
-9 - 15	Allowed range

Attribute <code>profile5-dBm</code>	
The threshold power level of peak power detector for profile5 in the steps of 0.1 dBm.	
-9 - 15	Allowed range

Attribute <code>profile6-dBm</code>	
The threshold power level of peak power detector for profile6 in the steps of 0.1 dBm.	
-9 - 15	Allowed range

Attribute <code>profile7-dBm</code>	
The threshold power level of peak power detector for profile7 in the steps of 0.1 dBm.	
-9 - 15	Allowed range

5.2.16 fusaFaultMask

The `fusaFaultMask` element configures the masking of the functional safety faults. The masked faults do not assert the `ERROR_N` signal.

Attribute <code>sr65-sr67-supply-low-1v8-tx2</code>	
TX2 supply low error (1V8)	

Attribute sr65-sr67-supply-low-1v8-tx2

0	Fault is unmasked
1	Fault is masked

Attribute sr65-sr67-supply-low-1v8-tx3

TX3 supply low error (1V8)

0	Fault is unmasked
1	Fault is masked

Attribute sr72-pr-tx1

Phase rotator error for TX1

0	Fault is unmasked
1	Fault is masked

Attribute sr72-pr-tx2

Phase rotator error for TX2

0	Fault is unmasked
1	Fault is masked

Attribute sr72-pr-tx3

Phase rotator error for TX3

0	Fault is unmasked
1	Fault is masked

Attribute sr65-sr67-supply-low-1v1-tx1

TX1 supply low error (1V1)

0	Fault is unmasked
1	Fault is masked

Attribute sr65-sr67-supply-low-1v1-tx2

TX2 supply low error (1V1)

0	Fault is unmasked
1	Fault is masked

Attribute sr65-sr67-supply-low-1v1-tx3

TX3 supply low error (1V1)

0	Fault is unmasked
1	Fault is masked

Attribute sr27-bb-tx3

RF connectivity to external antenna port error for TX3

0	Fault is unmasked
1	Fault is masked

Attribute sr69-rf-level-min-tx1

Min level monitoring error indicating TX1 RF level reached below min limit

0	Fault is unmasked
1	Fault is masked

Attribute sr69-rf-level-min-tx2

Min level monitoring error indicating TX2 RF level reached below min limit

0	Fault is unmasked
1	Fault is masked

Attribute sr69-rf-level-min-tx3

Min level monitoring error indicating TX3 RF level reached below min limit

0	Fault is unmasked
1	Fault is masked

Attribute sr69-rf-level-max-tx1

Max level monitoring error indicating TX1 RF level reached beyond max limit

0	Fault is unmasked
1	Fault is masked

Attribute sr69-rf-level-max-tx2

Max level monitoring error indicating TX2 RF level reached beyond max limit

0	Fault is unmasked
---	-------------------

Attribute sr69-rf-level-max-tx2

1	Fault is masked
---	-----------------

Attribute sr69-rf-level-max-tx3

Max level monitoring error indicating TX3 RF level reached beyond max limit

0	Fault is unmasked
1	Fault is masked

Attribute sr65-sr67-supply-low-1v8-tx1

TX1 supply low error (1V8)

0	Fault is unmasked
1	Fault is masked

Attribute sr65-sr67-supply-low-1v1-rx3

Supply 1v1 low error RX3

0	Fault is unmasked
1	Fault is masked

Attribute sr65-sr67-supply-low-1v1-rx4

Supply 1v1 low error RX4

0	Fault is unmasked
1	Fault is masked

Attribute sr65-sr67-supply-low-1v8-rx1

Supply 1v8 low error RX1

0	Fault is unmasked
1	Fault is masked

Attribute sr65-sr67-supply-low-1v8-rx2

Supply 1v8 low error RX2

0	Fault is unmasked
1	Fault is masked

Attribute sr65-sr67-supply-low-1v8-rx3

Supply 1v8 low error RX3

0	Fault is unmasked
1	Fault is masked

Attribute sr65-sr67-supply-low-1v8-rx4

Supply 1v8 low error RX4

0	Fault is unmasked
1	Fault is masked

Attribute sr27-bb-tx1

RF connectivity to external antenna port error for TX1

0	Fault is unmasked
1	Fault is masked

Attribute sr27-bb-tx2

RF connectivity to external antenna port error for TX1

0	Fault is unmasked
1	Fault is masked

Attribute s47-lo-level-min-rx3

Min Level monitoring error flag indicating RX3 LO level reached below min limit

0	Fault is unmasked
1	Fault is masked

Attribute s47-lo-level-min-rx4

Min Level monitoring error flag indicating RX4 LO level reached below min limit

0	Fault is unmasked
1	Fault is masked

Attribute s47-lo-level-max-rx1

Max Level monitoring error flag indicating RX1 LO level reached beyond max limit

0	Fault is unmasked
---	-------------------

Attribute s47-lo-level-max-rx1

1	Fault is masked
---	-----------------

Attribute s47-lo-level-max-rx2

Max Level monitoring error flag indicating RX2 LO level reached beyond max limit

0	Fault is unmasked
1	Fault is masked

Attribute s47-lo-level-max-rx3

Max Level monitoring error flag indicating RX3 LO level reached beyond max limit

0	Fault is unmasked
1	Fault is masked

Attribute s47-lo-level-max-rx4

Max Level monitoring error flag indicating RX4 LO level reached beyond max limit

0	Fault is unmasked
1	Fault is masked

Attribute sr65-sr67-supply-low-1v1-rx1

Supply 1v1 low error RX1

0	Fault is unmasked
1	Fault is masked

Attribute sr65-sr67-supply-low-1v1-rx2

Supply 1v1 low error RX2

0	Fault is unmasked
1	Fault is masked

Attribute sr46-aaFc-chirp

AAFC calibration error for chirp digital

0	Fault is unmasked
1	Fault is masked

Attribute sr46-aafe-cal-timeout-chirp

AAFC calibration timeout error for chirp digital

0	Fault is unmasked
1	Fault is masked

Attribute sr9-bb-rx1

RF connectivity to external antenna port error for RX1

0	Fault is unmasked
1	Fault is masked

Attribute sr9-bb-rx2

RF connectivity to external antenna port error for RX2

0	Fault is unmasked
1	Fault is masked

Attribute sr9-bb-rx3

RF connectivity to external antenna port error for RX3

0	Fault is unmasked
1	Fault is masked

Attribute sr9-bb-rx4

RF connectivity to external antenna port error for RX4

0	Fault is unmasked
1	Fault is masked

Attribute s47-lo-level-min-rx1

Min Level monitoring error flag indicating RX1 LO level reached below min limit

0	Fault is unmasked
1	Fault is masked

Attribute s47-lo-level-min-rx2

Min Level monitoring error flag indicating RX2 LO level reached below min limit

0	Fault is unmasked
---	-------------------

Attribute s47-lo-level-min-rx2

1	Fault is masked
---	-----------------

Attribute sr43-unlock-chirp

Error indicating chirp PLL in unlock condition

0	Fault is unmasked
1	Fault is masked

Attribute sr45-level-min-chirp

Min Level monitoring error flag indicating VCO output level reached below min limit

0	Fault is unmasked
1	Fault is masked

Attribute sr45-level-max-chirp

Max Level monitoring error flag indicating VCO output level reached beyond max limit

0	Fault is unmasked
1	Fault is masked

Attribute sr42-vco-freq-480-chirp

Error indicating VCO output frequency to the chirp digital logic is beyond programmable limit

0	Fault is unmasked
1	Fault is masked

Attribute sr7-interface-stuck-chirp

Stuck on chirp_seq_active, safety_monitoring, window_active detected

0	Fault is unmasked
1	Fault is masked

Attribute sr65-sr67-supply-low-pll-1v8-chirp

Supply 1V8 low error for chirp PLL

0	Fault is unmasked
1	Fault is masked

Attribute sr65-sr67-supply-low-vco-1v8-chirp

Supply 1V8 low error for chirp VCO

0	Fault is unmasked
1	Fault is masked

Attribute sr65-sr67-supply-low-1v1-chirp

Supply 1V1 low error for chirp digital

0	Fault is unmasked
1	Fault is masked

Attribute sr18-bb-lo-out-loi

RF connectivity issues in LO-out interface

0	Fault is unmasked
1	Fault is masked

Attribute sr17-bb-lo-in-loi

RF connectivity issues in LO-in interface

0	Fault is unmasked
1	Fault is masked

Attribute sr65-sr67-supply-low-1v1-loi

Supply low error from LO interface

0	Fault is unmasked
1	Fault is masked

Attribute sr19-wr-spi-crc

SPI CRC write error

0	Fault is unmasked
1	Fault is masked

Attribute sr19-rd-spi-crc

SPI CRC read error

0	Fault is unmasked
---	-------------------

Attribute sr19-rd-spi-crc

1	Fault is masked
---	-----------------

Attribute sr48-level-min-loi

Min Level monitoring error flag indicating LOI level reached below min limit

0	Fault is unmasked
1	Fault is masked

Attribute sr48-level-max-loi

Max Level monitoring error flag indicating LOI level reached beyond max limit

0	Fault is unmasked
1	Fault is masked

Attribute sr7-lock-step-chirp

Error indicating chirp digital logic in stuck condition or transient logic failure

0	Fault is unmasked
1	Fault is masked

Attribute sr61-supply-high-1v8-gldo

Global 1V8 LDO supply high sensor error

0	Fault is unmasked
1	Fault is masked

Attribute sr59-supply-low-1v1-gldo

Global 1V8 LDO supply low sensor error

0	Fault is unmasked
1	Fault is masked

Attribute sr57-supply-high-1v1-gldo

Global 1V8 LDO supply high sensor error

0	Fault is unmasked
1	Fault is masked

Attribute sr35-p11-lock-mclk

Error indicating P11 in unlock condition

0	Fault is unmasked
1	Fault is masked

Attribute sr39-dig-freq-mclk

Error indicating master clock digital output frequency not in range

0	Fault is unmasked
1	Fault is masked

Attribute sr65-sr67-supply-low-1v8-mclk

Supply error from master clock (1V8)

0	Fault is unmasked
1	Fault is masked

Attribute sr36-mspc-mclk

Leader-Follower phase calibration error

0	Fault is unmasked
1	Fault is masked

Attribute sr11-xo-no-clock-mclk

Clock failure (no clock error) from master clock

0	Fault is unmasked
1	Fault is masked

Attribute sr65-67-supply-low-1v8-adc12

Supply low (1v8) for ADC12

0	Fault is unmasked
1	Fault is masked

Attribute sr65-67-supply-low-1v8-adc34

Supply low (1v8) for ADC34

0	Fault is unmasked
---	-------------------

Attribute sr65-67-supply-low-1v8-adc34

1	Fault is masked
---	-----------------

Attribute sr65-67-supply-low-1v8-ssb

Supply 1V8 low sensor error for SSB MOD

0	Fault is unmasked
1	Fault is masked

Attribute sr65-67-supply-low-1v8-gbias

Supply 1V8 low sensor error for global bias

0	Fault is unmasked
1	Fault is masked

Attribute sr65-67-supply-low-1v1-ser

Supply 1V1 low sensor error for serializer digital and LVDS

0	Fault is unmasked
1	Fault is masked

Attribute sr65-67-supply-low-1v1-csi2

Supply 1V1 low sensor error for serializer CSI2

0	Fault is unmasked
1	Fault is masked

Attribute sr65-67-supply-low-1v8-ser

Supply 1V8 low sensor error for serializer

0	Fault is unmasked
1	Fault is masked

Attribute sr63-supply-low-1v8-gldo

Global 1V8 LDO supply low sensor error

0	Fault is unmasked
1	Fault is masked

Attribute sr31-bist-mixer-gaindiff-sw

RFBIST mixer gain test error

0	Fault is unmasked
1	Fault is masked

Attribute sr31-bist-lna-phasediff-sw

RFBIST LNA phase test error

0	Fault is unmasked
1	Fault is masked

Attribute sr31-bist-lna-gaindiff-sw

RFBIST LNA gain test error

0	Fault is unmasked
1	Fault is masked

Attribute sr3-reg-crc

Resgiter CRC error

0	Fault is unmasked
1	Fault is masked

Attribute sr34-cal-adc1

Calibration error for ADC1

0	Fault is unmasked
1	Fault is masked

Attribute sr34-cal-adc2

Calibration error for ADC2

0	Fault is unmasked
1	Fault is masked

Attribute sr34-cal-adc3

Calibration error for ADC3

0	Fault is unmasked
---	-------------------

Attribute sr34-cal-adc3

1	Fault is masked
---	-----------------

Attribute sr34-cal-adc4

Calibration error for ADC4

0	Fault is unmasked
1	Fault is masked

Attribute sr72-tx2-pr-cal-sw

TX2 phase rotator calibration check error

0	Fault is unmasked
1	Fault is masked

Attribute sr72-tx1-pr-cal-sw

TX1 phase rotator calibration check error

0	Fault is unmasked
1	Fault is masked

Attribute sr72-tx-pr-cal-sw

TX phase rotator calibration check error

0	Fault is unmasked
1	Fault is masked

Attribute sr70-tx-integrity-sw

TX integrity check error

0	Fault is unmasked
1	Fault is masked

Attribute sr44-cafc-integrity-sw

CAFC integrity check error

0	Fault is unmasked
1	Fault is masked

Attribute sr44-cafc-intgerity-unlock-sw

CAFC unlock detector integrity check error

0	Fault is unmasked
1	Fault is masked

Attribute sr44-cafc-intgerity-level-sw

CAFC level detector integrity check error

0	Fault is unmasked
1	Fault is masked

Attribute sr31-bist-mixer-phasediff-sw

RFBIST mixer phase test error

0	Fault is unmasked
1	Fault is masked

Attribute rfe-generic-sw

SW generic error

0	Fault is unmasked
1	Fault is masked

Attribute sr13-input-mode-toggle-test-sw

Input mode toggle test

0	Fault is unmasked
1	Fault is masked

Attribute sr14-disable-lb-toggle-test-sw

Disable loopback toggle test

0	Fault is unmasked
1	Fault is masked

Attribute sr72-tx3-pr-cal-sw

SPI CRC Read error.

0	Fault is unmasked
---	-------------------

Attribute <code>sr72-tx3-pr-cal-sw</code>	
1	Fault is masked

5.3 radarCycleConfig

The structure of the `radarCycleConfig` element is:

Element:	Occurence:
<code><radarCycleConfig></code>	1
<code><radarCycleDuration ...></code>	1
<code><chirpSequences ...></code>	1
<code><recalibration></code>	1
<code><recalibrationProfileIndependent ...></code>	0 - 1
<code><recalibrationProfileDependent ...></code>	0 - 8
<code></recalibration></code>	
<code><bistInterval ...></code>	1
<code></radarCycleConfig></code>	

5.3.1 radarCycleDuration

The `radarCycleDuration` element configures the duration of the radar cycle.

Attribute <code>ticks</code>	
Sets the duration of the radar cycle.	
400000 - 20000000	Radar cycle duration in ticks of 25 nanoseconds (e.g., 400000 ticks = 10 milliseconds).

5.3.2 chirpSequences

The `chirpSequences` element configures the chirp sequences.

Attribute <code>chirp-sequence-config-index</code>	
Determines the order of the chirp sequence of the radar cycle	
e.g., "0 3 4"	A space delimited list of <code>chirpSequenceConfig</code> indexes (0 - 7) that determine the chirp sequence of the radar cycle. This list must contain between 1 and 8 items.

Attribute <code>start-time-offset</code>	
The chirp sequence time offset in 25 nanoseconds resolution with respect to the start time of the radar cycle. The offset is defined as the time difference between the start of the dwell time of the first chirp (for each Sequence) and the start time of the radar cycle. This time offset can only be set for chirp sequence 0-7. The start time of chirp sequence 0 is fixed at design time. It depends on the timing of several states that the system has to go through, even though the value can be set to "0".	
e.g., "0 100 200 300"	A space delimited list of chirp sequence time offsets in ticks of 25 nanoseconds (0 - 20000000).

5.3.3 recalibrationProfileIndependent (not available)

The `recalibrationProfileIndependent` element defines before which chirp sequences to perform the chirp profile independent recalibrations (in the calibration phase). The chirp profile independent recalibrations should be performed regularly, e.g. once in the radar cycle.

Attribute <code>chirp-sequences</code>	
Defines before which chirp sequences to perform the chirp profile independent recalibrations (in the calibration phase).	
e.g., "0 3 4"	A space delimited list of chirp sequence indexes (0 – 7). This list must contain between 1 and 8 items and cannot contain duplicate values.

5.3.4 recalibrationProfileDependent (not available)

The `recalibrationProfileDependent` element selects the chirp profiles to recalibrate in the calibration phase before each chirp sequence. For each chirp sequence a different selection of chirp profiles can be set. Each chirp profile that is used in a chirp sequence should be recalibrated regularly, e.g. before every chirp sequence that uses that chirp profile.

Attribute <code>chirp-sequence</code>	
Defines before which chirp sequences to perform the chirp profile independent recalibrations (in the calibration phase).	
0 – 7	The chirp sequence index. Each <code>recalibrationProfileIndependent</code> elements in the <code>recalibration</code> element group must use a unique index.

Attribute <code>chirp-profiles</code>	
Defines before which chirp sequences to perform the chirp profile independent recalibrations (in the calibration phase).	
e.g., "0 3 4"	A space delimited list of chirp profile indexes (0 – 7). This list must contain between 1 and 8 items and cannot contain duplicate values.

5.3.5 bistInterval

The `bistInterval` element configures the interval to perform RFE built-in-self-tests.

Attribute <code>interval</code>	
Sets the interval to perform RFE built-in-self-tests.	
no bist	BIST is disabled.
every radar cycle	Perform BIST after the last chirp sequence in every radar cycle.
once in two radar cycles	Perform BIST after the last chirp sequence once in two radar cycles.
once in four radar cycles	Perform BIST after the last chirp sequence once in four radar cycles.

5.4 chirpSequenceConfigs

The structure of the `chirpSequenceConfigs` element is:

Element:	Occurence:
<chirpSequenceConfigs>	1
<chirpSequenceConfig>	1 - 8

Element:	Occurence:
<code><chirpCount ...></code>	1
<code><chirpProfileSequence ...></code>	1
<code><profileRepeatCount ...></code>	1
<code><dynamicUpdate ...></code>	1
<code><ddma ...></code>	1
<code><rxEnable ...></code>	1
<code><pdcc ...></code>	1
<code><txCalibrationThresholds ...></code>	1
<code></chirpSequenceConfig></code>	
<code></chirpSequenceConfigs></code>	

5.4.1 chirpCount

The `chirpCount` element configures the number of chirps within the chirp sequence.

Attribute count	
Sets the number of chirps within the chirp sequence.	
1 - 4096	Allowed range.

5.4.2 chirpProfileSequence

The `chirpProfileSequence` element configures the profile sequence that is repetitively applied to all the chirps in the chirp sequence.

Attribute sequence	
Sets the profile sequence that is repetitively applied to all the chirps in the chirp sequence.	
e.g., "0 3 4"	A space delimited list of chirp profile indexes (0 - 7). This list must contain between 1 and 8 items and cannot contain duplicate values.

5.4.3 profileRepeatCount

The `profileRepeatCount` element configures the number of times a profile will be repeated within the chirp sequence, before switching to the next profile in the list (e.g. using `profileRepeatCount` = 3, and `chirpProfileSequence` = "0 1 2" → 0, 0, 0, 1, 1, 1, 2, 2, 2. Otherwise, with `profileRepeatCount` = 1 → 0, 1, 2, 0, 1, 2, 0, 1, 2).

Attribute count	
Repeats the chirp using the same profile for a number of times, then switches to the next profile.	
1 - 127	Allowed range.

5.4.4 dynamicUpdate

The `dynamicUpdate` element configures the usage of dynamic tables.

Attribute <code>dynamic-updates</code>	
Disables or enables the dynamic update (i.e., parameter updates per chirp).	
<code>disabled</code>	Dynamic update is disabled.
<code>enabled</code>	Dynamic update is enabled.

Attribute <code>dynamic-table-index</code>	
The index of the dynamic table to be used for the chirp sequence. This table contains the values for each dynamic parameter for each chirp. The dynamic tables are to be created at design time.	
<code>0 - 7</code>	Allowed range.

5.4.5 txCalibrationThresholds (optional)

The `txCalibrationThresholds` element configures the high/low thresholds that would trigger the TX antenna power to be adjusted automatically during calibration. The increments are done in steps of 0.29dB, however any value can be used from the ranges below, and the actual value that will end up on the front-end will automatically be set to the closest mutiple of 0.29dB relative to the value used in XML.

Note: This parameter is only applicable if `txCalibration` ([Section 5.1.7](#)) is also set to `without transmission`

Example: `xml value` = 1.27dB → **front-end value** = 1.16dB (closest mutiple of 0.29dB)

Attribute <code>tx-rf-threshold-high-dB</code>	
Configures the threshold that once reached, would cause the TX antenna output power to be decreased.	
<code>0.01 - 4</code>	Allowed range.
<code>2</code>	Default value (when parameter omitted from XML).

Attribute <code>tx-rf-threshold-low-dB</code>	
Configures the threshold that once reached, would cause the TX antenna output power to be increased.	
<code>0.01 - 4</code>	Allowed range.
<code>3</code>	Default value (when parameter omitted from XML).

5.4.6 ddma

The `ddma` element configures the DDMA ramp generator settings.

Attribute <code>mode</code>	
Configures the DDMA control mode.	
<code>disabled</code>	DDMA is disabled
<code>fun mode</code>	DDMA functional mode (phase code step incremented every chirp)
<code>test mode</code>	DDMA test mode (phase code step is incremented during chirp @ 40 MHz)

Attribute <code>mode</code>	
test no chirp	DDMA test mode with non-chirp mode (phase code step is incremented during chirp @ 40 MHz)

Attribute <code>tx-1-use-ddma</code>	
Configures the use of DDMA generator for TX 1.	
false	Phase code is taken from Profile..
true	Phase code is taken from DDMA generator.

Attribute <code>tx-2-use-ddma</code>	
Configures the use of DDMA generator for TX 2.	
false	Phase code is taken from Profile.
true	Phase code is taken from DDMA generator.

Attribute <code>tx-3-use-ddma</code>	
Configures the use of DDMA generator for TX 3.	
false	Phase code is taken from Profile.
true	Phase code is taken from DDMA generator.

Attribute <code>tx-1-ddma-init-phase</code>	
Configures the PR DDMA initial phase for TX 1.	
0 - 354.375	Phase rotation in degrees (in steps of 5.625 degrees).

Attribute <code>tx-2-ddma-init-phase</code>	
Configures the PR DDMA initial phase for TX 2.	
0 - 354.375	Phase rotation in degrees (in steps of 5.625 degrees).

Attribute <code>tx-3-ddma-init-phase</code>	
Configures the PR DDMA initial phase for TX 3.	
0 - 354.375	Phase rotation in degrees (in steps of 5.625 degrees).

Attribute <code>tx-1-phase-update</code>	
Configures the PR DDMA phase increment for TX 1.	

Attribute tx-1-phase-update	
0 - 354.375	Phase rotation in degrees (in steps of 5.625 degrees).

Attribute tx-2-phase-update	
Configures the PR DDMA phase increment for TX 2.	
0 - 354.375	Phase rotation in degrees (in steps of 5.625 degrees).

Attribute tx-3-phase-update	
Configures the PR DDMA phase increment for TX 3.	
0 - 354.375	Phase rotation in degrees (in steps of 5.625 degrees).

Attribute tx-1-final-pc-gen-mode	
Configures how the phase code is generated for TX 1.	
no change	No change to phase code.
replace msb with qpsk	Replace two MSB bits of phase code with the QPSK[1:0] bits
add qpsk to msb	Add the QPSK[1:0] bits to two MSB bits of initial phase code

Attribute tx-2-final-pc-gen-mode	
Configures how the phase code is generated for TX 2.	
no change	No change to phase code.
replace msb with qpsk	Replace two MSB bits of phase code with the QPSK[1:0] bits
add qpsk to msb	Add the QPSK[1:0] bits to two MSB bits of initial phase code

Attribute tx-3-final-pc-gen-mode	
Configures how the phase code is generated for TX 3.	
no change	No change to phase code.
replace msb with qpsk	Replace two MSB bits of phase code with the QPSK[1:0] bits
add qpsk to msb	Add the QPSK[1:0] bits to two MSB bits of initial phase code

5.4.7 rxEnable

The `rxEnable` element configures the state of the receivers during the chirp.

Attribute rx-1	
Disables or enables receiver 1.	
disabled	Receiver 1 is disabled during the chirp.

Attribute rx-1	
enabled	Receiver 1 is enabled during the chirp.

Attribute rx-2	
Disables or enables receiver 2.	
disabled	Receiver 2 is disabled during the chirp.
enabled	Receiver 2 is enabled during the chirp.

Attribute rx-3	
Disables or enables receiver 3.	
disabled	Receiver 3 is disabled during the chirp.
enabled	Receiver 3 is enabled during the chirp.

Attribute rx-4	
Disables or enables receiver 4.	
disabled	Receiver 4 is disabled during the chirp.
enabled	Receiver 4 is enabled during the chirp.

5.4.8 pdc (not available)

The `pdc` element configures the programmable decimation chain of the RFE.

Attribute notch-filter-enable	
Disables or enables the DC notch filter.	
false	The DC notch filter is disabled.
true	The DC notch filter is enabled.

5.5 chirpProfiles

The structure of the `chirpProfiles` element is:

Element:	Occurence:
<chirpProfiles>	1
<chirpProfiles>	1 - 8
<effectiveSamplingFrequency ...>	1
<chirpTiming ...>	1
<chirpGenerator ...>	1
<txTransmissionEnable ...>	1
<txPower ...>	1

Element:	Occurence:
<txPhaseRotation ...>	1
<chirpFrequency ...>	1
<rxGain ...>	1
<rxFilter ...>	1
<csi2VirtualChannel ...>	1
</chirpProfiles>	
</chirpProfiles>	

5.5.1 effectiveSamplingFrequency

The `effectiveSamplingFrequency` element configures the effective sampling frequency of the chirp. Here 'effective' indicates that this is the sampling frequency after decimation, i.e. of the output samples.

Attribute frequency	
Sets the effective sampling frequency of the chirp.	
5 MHz	The effective sampling frequency is set to 5 MHz.
10 MHz	The effective sampling frequency is set to 10 MHz.
20 MHz	The effective sampling frequency is set to 20 MHz.
40 MHz	The effective sampling frequency is set to 40 MHz.

5.5.2 chirpTiming

The `chirpTiming` element configures the timing of the chirp.

Note: There is a timing constraint internal to TEF82xx called **T_{return}** = dwell-time-ticks + settle-time-ticks + reset-time-ticks. T_{return} has a minimum required value of **160 ticks**, which means that setting dwell, settle and reset times to their minimum allowed value, will result in an error ($40 + 80 + 30 < 160$). The solution is to increase the value of either of the 3 components of T_{return} in order to fulfill the restriction.

Attribute chirp-interval-time-ticks	
Sets the chirp interval time in 25 nanoseconds resolution. The <i>chirp interval time</i> is the time between the start of subsequent chirps, and this parameter can be used to introduce additional idle time between chirps. The value for this parameter must be equal or larger than the sum of all its components: dwell-time-ticks, settle-time-ticks, acquisition-time-ticks, jumpbackTiming (Section 5.1.6), and reset-time-ticks.	
400 - 4194303	Allowed range.

Attribute dwell-time-ticks	
Sets the chirp dwell time in 25 nanoseconds resolution.	
40 - 4094	Allowed range.

Attribute settle-time-ticks	
Sets the chirp settle time in 25 nanoseconds resolution.	

Attribute `settle-time-ticks`

80 - 4094 Allowed range.

Attribute `acquisition-time-ticks`

Sets the chirp acquisition time in 25 nanoseconds resolution.

8 - 8192 Allowed range.

Attribute `reset-time-ticks`

Sets the chirp reset time in 25 nanoseconds resolution.

30 - 4094 Allowed range.

5.5.3 chirpGenerator

The `chirpGenerator` element configures the chirp generator parameters.

Attribute `center-frequency-kHz`

Sets the center frequency of the chirp in kHz.

Note: If Frequency Drift is enabled ([Section 5.5.7](#)), then using profiles with different Center Frequency in the same sequence will result in only the first Center Frequency being used, and the rest will be ignored.

The tool will output a warning if it detects such a configuration.

76000000 - 81000000 Allowed range.

Attribute `effective-chirp-bandwidth-kHz`

Sets the effective bandwidth in kHz. This is the bandwidth used during the chirp acquisition. A bandwidth of 0 results in a 'chirp' with constant frequency.

0 - 4000000 Allowed range.

Attribute `vco`

Configures the VCO bandwidth that the chirp PLL uses to generate the chirp.

1 GHz	The 1 GHz VCO is selected.
2 GHz	The 2 GHz VCO is selected.
4 GHz	The 4 GHz VCO is selected.

Attribute `slope`

Selects the direction of the chirp slope.

falling	A falling slope is selected.
rising	A rising slope is selected.

Attribute pll-loop-filter-bandwidth	
Selects the chirp PLL loop filter bandwidth. Values that can be selected, depend on the <code>vco</code> parameter from above. In order to achieve the highest recommended chirp steepness, the values below are suggested, which also ensure the lowest phase noise: • For 1GHz VCO → 200kHz 250kHz 300kHz • For 2GHz VCO → 250kHz 300kHz • For 4GHz VCO → 300kHz 350kHz Note: If using values that are higher than the ones recommended in the above list for each VCO category, the tool will output a warning, notifying the user of the increased phase noise, however the blob will still be generated successfully.	
200 kHz	The PLL loop filter bandwidth is 200 kHz. <i>(Valid only for 1GHz VCO)</i>
250 kHz	The PLL loop filter bandwidth is 250 kHz. <i>(Valid only for 1GHz & 2GHz VCO)</i>
300 kHz	The PLL loop filter bandwidth is 300 kHz.
350 kHz	The PLL loop filter bandwidth is 350 kHz.
400 kHz	The PLL loop filter bandwidth is 400 kHz.
450 kHz	The PLL loop filter bandwidth is 450 kHz.
500 kHz	The PLL loop filter bandwidth is 500 kHz.
550 kHz	The PLL loop filter bandwidth is 550 kHz.
600 kHz	The PLL loop filter bandwidth is 600 kHz.
650 kHz	The PLL loop filter bandwidth is 650 kHz.
700 kHz	The PLL loop filter bandwidth is 700 kHz.
750 kHz	The PLL loop filter bandwidth is 750 kHz.
800 kHz	The PLL loop filter bandwidth is 800 kHz.
850 kHz	The PLL loop filter bandwidth is 850 kHz.
900 kHz	The PLL loop filter bandwidth is 900 kHz.
950 kHz	The PLL loop filter bandwidth is 950 kHz.
1000 kHz	The PLL loop filter bandwidth is 1000 kHz.
1050 kHz	The PLL loop filter bandwidth is 1050 kHz.
1100 kHz	The PLL loop filter bandwidth is 1100 kHz.
1150 kHz	The PLL loop filter bandwidth is 1150 kHz.
1200 kHz	The PLL loop filter bandwidth is 1200 kHz.
1250 kHz	The PLL loop filter bandwidth is 1250 kHz.
1300 kHz	The PLL loop filter bandwidth is 1300 kHz.
1350 kHz	The PLL loop filter bandwidth is 1350 kHz.
1400 kHz	The PLL loop filter bandwidth is 1400 kHz.
1450 kHz	The PLL loop filter bandwidth is 1450 kHz.
1500 kHz	The PLL loop filter bandwidth is 1500 kHz.
1550 kHz	The PLL loop filter bandwidth is 1550 kHz.
1600 kHz	The PLL loop filter bandwidth is 1600 kHz.
1650 kHz	The PLL loop filter bandwidth is 1650 kHz. <i>(Valid only for 2GHz & 4GHz VCO)</i>
1700 kHz	The PLL loop filter bandwidth is 1700 kHz. <i>(Valid only for 4GHz VCO)</i>

5.5.4 txTransmissionEnable

The txTransmissionEnable element configures the enabling of TX antennas. If a Sequence uses multiple profiles, that has different TXs enabled, then power mode ([Section 5.1.1](#)) must be configured to "lower power" otherwise a lock-step error would be triggered from ISM. To prevent this, the generator will not generate the blob files if "lower power" is not used in the described scenario, and it will display the error message: *"Attempting to use different TX antennas per profile, without setting <power mode= "lower power"> "*

Attribute tx-1	
Configure the transmission enabled setting for transmitter 1.	
disabled	Transmitter 1 is not transmitting during the chirp.
enabled	Transmitter 1 is transmitting during the chirp.

Attribute tx-2	
Configure the transmission enabled setting for transmitter 2.	
disabled	Transmitter 2 is not transmitting during the chirp.
enabled	Transmitter 2 is transmitting during the chirp.

Attribute tx-3	
Configure the transmission enabled setting for transmitter 3.	
disabled	Transmitter 3 is not transmitting during the chirp.
enabled	Transmitter 3 is transmitting during the chirp.

Attribute time-reference (not available)	
Sets the reference time to enable the transmission and apply the phase rotation after a fixed delay.	
start-of-dwell-time	The start of the dwell time is used as the reference.
start-of-settle-time	The start of the settle time is used as the reference.

Attribute time-offset-ticks (not available)	
Sets the time offset relative to the reference time (see attribute time-reference) after which the transmission is enabled and the phase rotation applied.	
2 - 4094	Offset is in 25 nanosenconds resolution.

5.5.5 txPower

The txPower element configures the output power of the transmitters.

Attribute dBm	
Sets the output power of the transmitters in steps of 0.1 dBm.	

Attribute dBm	
3 - 15	Transmitter power in dBm.

5.5.6 txPhaseRotation

The txPhaseRotation element configures the phase rotation for each transmitter.

Attribute tx-1	
Sets the phase rotation for transmitter 1.	
0 - 354.375	Phase rotation in degrees (in steps of 5.625 degrees).

Attribute tx-2	
Sets the phase rotation for transmitter 2.	
0 - 354.375	Phase rotation in degrees (in steps of 5.625 degrees).

Attribute tx-3	
Sets the phase rotation for transmitter 3.	
0 - 354.375	Phase rotation in degrees (in steps of 5.625 degrees).

5.5.7 chirpFrequency

The chirpFrequency element configures the chirp frequency parameters.

Attribute frequency-drift-steps	
The frequency of the chirp (i.e., start & end frequency of the chirp frequency sweep) increments (positive) or decrements (negative) by x * 42.9153 Hz each chirp. A value of 0 will disable the drift. Values other than 0 are not supported, when the chirp sequence consists of chirps with different Center Frequencies (i.e. center-frequency-kHz) (Section 5.5.3).	
0 - 932068	Allowed range.

Example: frequency-drift-steps="5" (XML) means that every chirp (after the first chirp) will have the start frequency increased by 5 * 42.9153 = 214.5765 Hz.

5.5.8 rxGain

The rxGain element configures the gain of the receivers.

Attribute gain	
Sets the gain of the receivers.	
27 dB	The receiver gain is set to 27 dB.
30 dB	The receiver gain is set to 30 dB.
33 dB	The receiver gain is set to 33 dB.
36 dB	The receiver gain is set to 36 dB.
39 dB	The receiver gain is set to 39 dB.

Attribute gain	
42 dB	The receiver gain is set to 42 dB.
45 dB	The receiver gain is set to 45 dB.

5.5.9 rxFilter

The rxFilter element configures the high and low pass filters of the receivers.

Attribute high-pass-filter	
Sets the high pass filter of the receivers.	
100 kHz	The high pass filter is set to 100 kHz.
200 kHz	The high pass filter is set to 200 kHz.
300 kHz	The high pass filter is set to 300 kHz.
400 kHz	The high pass filter is set to 400 kHz.
800 kHz	The high pass filter is set to 800 kHz.
1600 kHz	The high pass filter is set to 1600 kHz.
3200 kHz	The high pass filter is set to 3200 kHz.

Attribute low-pass-filter	
Sets the low pass filter of the receivers.	
12P5 MHz	The low pass filter is set to 12.5 MHz.
15 MHz	The low pass filter is set to 15 MHz.
20 MHz	The low pass filter is set to 20 MHz.
25 MHz	The low pass filter is set to 25 MHz.
40 MHz	The low pass filter is set to 40 MHz.

5.5.10 csi2VirtualChannel

The csi2VirtualChannel element configures the virtual channel applied on data for the chirps using this profile.

Attribute channel	
The virtual channel applied on data for the chirps using this profile.	
0	Virtual channel 0 is selected.
1	Virtual channel 1 is selected.
2	Virtual channel 2 is selected.

5.6 dynamicTables

The structure of the dynamicTables element is:

Element:	Occurence:
<dynamicTables>	0 - 1
<dynamicTable>	1 - 256
<dynamicTableEntry ...>	1 - 32768
</dynamicTable>	
</dynamicTables>	

5.6.1 dynamicTableEntry

The `dynamicTableEntry` element is used to update the phase rotation of each TX. The update is performed each chirp, in the order specified by the user, with each entry (some examples are shown below). The way in which a `dynamicTable` affects the Phase Rotation of TX antennas is dependent on attribute `tx-1-final-pc-gen-mode` (and their respective TX2/TX3 attributes) from (Section 5.4.6). Each of the values used (0, 90, 180, 270 degrees) corresponds with an I & Q value on the QPSK I/O lines, and whether these values *replace* OR *are added to* the 2 MSB of the 6 bit phase rotation values, are dependent on the `tx-[1/2/3]-final-pc-gen-mode` attribute mentioned earlier.

```
<dynamicTable>
  <dynamicTableEntry phase-rotation-tx-1="0"    phase-rotation-tx-2="270" phase-rotation-tx-3="0"/>
  <dynamicTableEntry phase-rotation-tx-1="90"   phase-rotation-tx-2="270" phase-rotation-tx-3="0"/>
  <dynamicTableEntry phase-rotation-tx-1="180"  phase-rotation-tx-2="180" phase-rotation-tx-3="90"/>
  <dynamicTableEntry phase-rotation-tx-1="270"  phase-rotation-tx-2="180" phase-rotation-tx-3="90"/>
  <dynamicTableEntry phase-rotation-tx-1="0"    phase-rotation-tx-2="90"  phase-rotation-tx-3="180"/>
  <dynamicTableEntry phase-rotation-tx-1="90"   phase-rotation-tx-2="90"  phase-rotation-tx-3="180"/>
  <dynamicTableEntry phase-rotation-tx-1="180"  phase-rotation-tx-2="0"   phase-rotation-tx-3="270"/>
  <dynamicTableEntry phase-rotation-tx-1="270"  phase-rotation-tx-2="0"   phase-rotation-tx-3="270"/>
</dynamicTable>
```

In cases where the number of chirps in a sequence (see Section 5.4.1) is larger than the number of dynamic table entries, the TEF82xx RFE firmware will continuously loop over the available dynamic table entries. This allows for a reduction in size of the generated dynamic table for chirp sequences that contain periodic values.

For example, the following `dynamicTables` element in a `rfeConfig.xml` file will generate a dynamic table with 4 entries for `phase-rotation-tx-1`, `phase-rotation-tx-2` and `phase-rotation-tx-3`:

```
<dynamicTable>
  <dynamicTableEntry phase-rotation-tx-1="0"    phase-rotation-tx-2="180" phase-rotation-tx-3="270"/>
  <dynamicTableEntry phase-rotation-tx-1="90"   phase-rotation-tx-2="0"   phase-rotation-tx-3="270"/>
  <dynamicTableEntry phase-rotation-tx-1="180"  phase-rotation-tx-2="180" phase-rotation-tx-3="90"/>
  <dynamicTableEntry phase-rotation-tx-1="270"  phase-rotation-tx-2="0"   phase-rotation-tx-3="90"/>
</dynamicTable>
```

The following phase rotations will be applied by the RFE firmware to TX 1, TX 2 & TX 3 in case the `chirpCount` is set to 4:

TX:	Chirp index:							
	1	2	3	4				
TX 1	0	90	180	270				
TX 2	180	0	180	0				
TX 3	270	270	90	90				

The following phase rotations will be applied by the RFE firmware to TX 1, TX 2 & TX 3 in case the `chirpCount` is set to 6:

TX:	Chirp index:							
	1	2	3	4	5	6		
TX 1	0	90	180	270	0	90		
TX 2	180	0	180	0	180	0		
TX 3	270	270	90	90	270	270		

The following phase rotations will be applied by the RFE firmware to TX 1, TX 2 & TX 3 in case the chirpCount is set to 8:

TX:	Chirp index:							
	1	2	3	4	5	6	7	8
TX 1	0	90	180	270	0	90	180	270
TX 2	180	0	180	0	180	0	180	0
TX 3	270	270	90	90	270	270	90	90

The available parameters that can be changed with dynamic tables are:

Attribute phase-rotation-tx-1	
Sets the phase rotation for transmitter 1.	
0	Phase rotation of 0 degrees.
90	Phase rotation of 90 degrees.
180	Phase rotation of 180 degrees.
270	Phase rotation of 270 degrees.

Attribute phase-rotation-tx-2	
Sets the phase rotation for transmitter 2.	
0	Phase rotation of 0 degrees.
90	Phase rotation of 90 degrees.
180	Phase rotation of 180 degrees.
270	Phase rotation of 270 degrees.

Attribute phase-rotation-tx-3	
Sets the phase rotation for transmitter 3.	
0	Phase rotation of 0 degrees.
90	Phase rotation of 90 degrees.
180	Phase rotation of 180 degrees.
270	Phase rotation of 270 degrees.

Note: The dynamic table blob is dynamically generated, i.e. the more entries you add in the XML the bigger the dynamic table blob gets. The only limitation is that the table should be under 40 KB. The structure of the data is

explained in the tables below (little endian). The order of the table elements is the same order of the data in the Dynamic Table blob.

Dynamic Table structure	
Data	Size
Total blob size (in Bytes)	2 Bytes
Number of tables (<i>N</i>)	1 Byte
Table header 0	4 Bytes
Table header 1	4 Bytes
...	...
Table header (<i>N</i> -1)	4 Bytes
Table data 0	1 Byte
Table data 1	1 Byte
...	...
Table data (<i>N</i> -1)	1 Byte

, where phase rotation is encoded in “Table data”, as follows:

Phase Rotation encoding for single TX (Table data)	
phase-rotation-tx-1	Byte encoding (1 Byte)
0	0000 0000
90	0000 0001
180	0000 0010
270	0000 0011

, and where

TX placement in Phase Rotation encoding (Table data)	
TX number	Bit placement (1 Byte)
phase-rotation-tx-1	0000 0000
phase-rotation-tx-2	0000 0001
phase-rotation-tx-3	0000 0010
reserved	0000 0011

Also, the “Table header”, is composed of the following data:

Table data encoding	
Data	Size
Data offset index (in Bytes)	2 Bytes
Data size (in Bytes)	2 Bytes

Note: “Data offset index (in Bytes)” from the “Dynamic Table header structure” table, starts counting from the absolute start of the data array.

6 Radar Cycle Schedule Generation

The RFE FW determines the radar cycle schedule, from the RFE configuration, as provided via `rfe_configure()` of the RFE Abstract API. The user can also derive this radar cycle schedule by using the **generateRadarCycleSchedule.exe** tool. This executable needs to be generated, using the available script in the `[rsdk_package]/RFE_abstract/RFE_driver2/CfgGen_RadarCycleSchedule/buildExecutable.bat`. Instructions on how to generate and use the executable, are available in the readme file found in the same location. When the input XML configuration file is valid, the tool will output a `radarCycleSchedule.csv` file. This file describes the start and end time in microseconds for each RFE radar cycle activity. It also lists the corresponding RFE state that can be observed via `rfe_getState()` of the RFE Abstract API. Generation of this schedule will return an error when the given RFE configuration is invalid. The returned error codes correspond to the errors listed in `rfe_error.h` of the RFE Abstract API.

Note: Starting with version **0.8.10**, this executable will also issue an error if any of the Sequences do not have enough idle time. The minimum idle time for each Sequence is **50µS**, which ensures enough time to process a “Stop” command. For this reason, it is highly recommended that the executable is called after generating the blob, to validate the configuration against timing restrictions.

	A	B	C	D
1	Activity	Start (us)	End (us)	State
2	Radar Cycle Begin	0.000	15.000	rfe_busy_e
3				
4	Chirp Sequence Init 0	15.000	115.000	rfe_busy_e
5	Power On 0	115.000	1115.000	rfe_busy_e
6	Recalibration 0	1115.000	4395.000	rfe_busy_e
7	Chirp Sequence Start 0	4395.000	4470.000	rfe_busy_e
8	Chirp Sequence 0	4470.000	9590.000	rfe_busy_e
9	Chirp Sequence End 0	9590.000	9665.000	rfe_busy_e
10	Monitors and Metadata 0	9665.000	9765.000	rfe_busy_e
11	Standby 0	9765.000	9965.000	rfe_busy_e
12	Radar Cycle Idle 0	9965.000	23075.000	rfe_radarCycleIdle_e
13	Command Buffer 0	23075.000	23095.000	rfe_busy_e
14				
15	Chirp Sequence Init 1	23095.000	23195.000	rfe_busy_e
16	Power On 1	23195.000	24195.000	rfe_busy_e
17	Recalibration 1	24195.000	24925.000	rfe_busy_e
18	Chirp Sequence Start 1	24925.000	25000.000	rfe_busy_e
19	Chirp Sequence 1	25000.000	30120.000	rfe_busy_e
20	Chirp Sequence End 1	30120.000	30195.000	rfe_busy_e
21	Monitors and Metadata 1	30195.000	30295.000	rfe_busy_e
22	BIST	30295.000	32295.000	rfe_busy_e
23	Standby 1	32295.000	32495.000	rfe_busy_e
24	Radar Cycle Idle 1	32495.000	49480.000	rfe_radarCycleIdle_e
25	Command Buffer 1	49480.000	49500.000	rfe_busy_e
26				
27	Config Update	49500.000	50000.000	rfe_busy_e

Figure 7. Radar cycle schedule csv file example

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